

ASSIGNMENT 3

SUBJECT CODE: CSU1658

SUBJECT NAME: Statistical Foundation of Data Sciences

TOTAL POINTS: 100

DEADLINE: DECEMBER 20TH 2025 4PM

K-Nearest Neighbors (KNN)

PROBLEM 1: Medical Diagnosis Classification

A hospital classifies patients based on glucose level (x_1) and BMI (x_2) as diabetic (+) or non-diabetic (-).

Training data:

Patient	Glucose	BMI	Class
<i>A</i>	85	22	—
<i>B</i>	95	25	—
<i>C</i>	140	32	+
<i>D</i>	160	35	+
<i>E</i>	120	28	+
<i>F</i>	90	24	—

New patient: $P = (110, 27)$.

- (a) Compute Euclidean distances from P to all training patients. Normalize features first using $\tilde{x} = (x - \min)/(max - \min)$.
- (b) Classify using $k = 3$ with majority voting. Show the complete decision process.
- (c) Derive the decision boundary equation for 1-NN between two nearest neighbors algebraically.

PROBLEM 2: Curse of Dimensionality Analysis

Consider KNN in high-dimensional spaces.

- (a) Prove that in d dimensions, the volume of a hypersphere of radius r is $V_d(r) = \frac{\pi^{d/2}}{\Gamma(d/2+1)} r^d$. Show that as $d \rightarrow \infty$, most volume concentrates near the surface.

- (b) Given 1000 uniformly distributed points in $[0, 1]^d$, derive the expected distance to the nearest neighbor as a function of d . Calculate for $d = 2, 10, 50$.
- (c) Explain why KNN performance degrades in high dimensions. Propose two methods to mitigate this issue.

PROBLEM 3: Customer Segmentation with Manhattan Distance

E-commerce data: purchases per month (x_1) and average order value (x_2):

ID	Purchases	Avg Value	Segment
1	2	50	Low
2	8	120	High
3	5	80	Medium
4	12	200	High
5	3	60	Low
6	7	90	Medium

New customer: $C = (6, 100)$.

- (a) Calculate Manhattan distances: $d_M(x, y) = |x_1 - y_1| + |x_2 - y_2|$. Why might Manhattan be better than Euclidean here?
- (b) Classify C using weighted KNN with $k = 4$ and weights $w_i = e^{-d_i}$. Show all calculations.
- (c) Implement leave-one-out cross-validation to find optimal $k \in \{1, 3, 5\}$. Which k minimizes error?

PROBLEM 4: KNN Regression for Temperature Prediction

Weather stations record altitude (m) and temperature ($^{\circ}\text{C}$):

Station	Altitude	Temperature
S_1	100	28
S_2	500	22
S_3	1000	15
S_4	1500	10
S_5	2000	5

Predict temperature at altitude 1200m using KNN regression.

- (a) Use $k = 3$ with inverse distance weighting: $\hat{y} = \frac{\sum_{i=1}^k w_i y_i}{\sum_{i=1}^k w_i}$ where $w_i = 1/d_i^2$. Calculate the predicted temperature.
- (b) Derive the bias-variance decomposition for KNN regression. How does k affect bias and variance?
- (c) Prove that as $k \rightarrow n$, KNN regression converges to the global mean. What does this imply about underfitting?

PROBLEM 5: KD-Tree Construction and Complexity

Dataset in 2D: $\{(2, 3), (5, 4), (9, 6), (4, 7), (8, 1), (7, 2)\}$.

- (a) Construct a KD-tree by recursively splitting on alternating dimensions. Draw the tree structure and partitioning regions.
- (b) Search for the nearest neighbor of query point $(6, 5)$ using the KD-tree. Show the pruning decisions and nodes visited.
- (c) Prove that KD-tree query time is $O(\log n)$ on average for balanced trees. When does it degrade to $O(n)$?

K-Means Clustering

PROBLEM 6: Image Color Quantization

Image has 9 pixels with RGB values (simplified):

Pixel	R	G	B
1	255	0	0
2	250	10	5
3	245	5	10
4	0	255	0
5	5	250	10
6	10	245	5
7	0	0	255
8	5	10	250
9	10	5	245

- (a) Run K-means with $k = 3$ initialized at pixels 1, 4, 7. Show two complete iterations (assignment + update).
- (b) Calculate final WCSS and explain how this reduces image file size.
- (c) Derive the relationship between K-means and vector quantization in signal processing.

PROBLEM 7: K-Means++ Initialization

Consider points: $\{1, 4, 8, 12, 15, 18, 22\}$ in 1D, $k = 3$.

- (a) Implement K-means++ initialization: choose first centroid randomly, then select subsequent centroids with probability proportional to $D(x)^2$ (squared distance to nearest centroid). Show the probability distribution at each step.
- (b) Compare final WCSS from K-means++ vs random initialization $\{\mu_1 = 1, \mu_2 = 4, \mu_3 = 8\}$.
- (c) Prove that K-means++ gives an $O(\log k)$ -approximation to optimal clustering in expectation.

PROBLEM 8: Soft K-Means (Gaussian Mixture Model)

Dataset: $\{0.5, 1.5, 2.0, 8.5, 9.0, 9.5\}$, $k = 2$.

- (a) Derive the soft K-means (EM for GMM) update equations:

$$r_{ik} = \frac{\exp(-\beta \|x_i - \mu_k\|^2)}{\sum_j \exp(-\beta \|x_i - \mu_j\|^2)}$$

where $\beta = 1/(2\sigma^2)$.

- (b) Run one EM iteration with $\mu_1 = 1, \mu_2 = 9, \sigma^2 = 1$. Calculate responsibilities r_{ik} and update $\mu_k = \frac{\sum_i r_{ik} x_i}{\sum_i r_{ik}}$.
- (c) Compare hard vs soft clustering. When is soft clustering preferred?

PROBLEM 9: Spectral Clustering Foundation

Similarity graph with 5 nodes and adjacency matrix:

$$W = \begin{pmatrix} 0 & 8 & 2 & 0 & 0 \\ 8 & 0 & 7 & 0 & 0 \\ 2 & 7 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 9 \\ 0 & 0 & 1 & 9 & 0 \end{pmatrix}$$

- (a) Compute the degree matrix D and Laplacian $L = D - W$. Find eigenvalues and eigenvectors of L .
- (b) Use the Fiedler vector (second smallest eigenvalue's eigenvector) to partition the graph into 2 clusters.
- (c) Explain why spectral clustering can find non-convex clusters while K-means cannot. Give a geometric example.

PROBLEM 10: Hierarchical vs K-Means

Gene expression levels (simplified): $\{2.1, 2.5, 2.3, 7.8, 8.2, 7.5, 15.1, 15.5\}$.

- (a) Perform agglomerative hierarchical clustering with complete linkage. Draw the dendrogram.
- (b) Cut dendrogram to get 3 clusters. Run K-means with $k = 3$ (random init). Compare cluster assignments.
- (c) Prove that hierarchical clustering has time complexity $O(n^3)$ for complete linkage while K-means is $O(ikn)$. Discuss trade-offs.

Principal Component Analysis (PCA)

PROBLEM 11: Stock Portfolio Dimensionality Reduction

Returns of 3 stocks over 4 days (percentages):

$$X = \begin{pmatrix} 2 & 3 & 2.5 \\ 1 & 2 & 1.5 \\ -1 & -2 & -1.5 \\ -2 & -3 & -2.5 \end{pmatrix}$$

- (a) Center the data and compute covariance matrix $\Sigma = \frac{1}{n-1}X^T X$. Interpret Σ_{12} .
- (b) Find eigenvalues and eigenvectors. Calculate the first principal component direction and interpret it financially.
- (c) Project data onto PC1. What percentage of portfolio variance is explained by this single factor?

PROBLEM 12: PCA via SVD for Tall Matrix

Data matrix (8 observations, 2 features):

$$X = \begin{pmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 5 \\ 4 & 7 \\ 5 & 9 \\ 6 & 11 \\ 7 & 12 \\ 8 & 14 \end{pmatrix}$$

(assume centered)

- (a) Compute SVD: $X = U\Sigma V^T$. Show that principal components are columns of V .
- (b) Derive the relationship: eigenvalues of $X^T X$ equal σ_i^2 (squared singular values).
- (c) Calculate reconstruction error when approximating X with rank-1 matrix: $\|X - X_1\|_F^2$. Verify it equals σ_2^2 .

PROBLEM 13: Kernel PCA for Nonlinear Data

1D data: $X = \{-3, -1, 0, 1, 3\}$ with nonlinear relationship.

- (a) Map to 2D using $\phi(x) = (x, x^2)$. Compute the mapped data matrix and its covariance.
- (b) Derive the kernel matrix $K_{ij} = \phi(x_i)^T \phi(x_j) = x_i x_j + x_i^2 x_j^2$ without explicit feature mapping.
- (c) Perform kernel PCA by eigendecomposition of centered kernel matrix. Compare with standard PCA on original data.

PROBLEM 14: PCA for Face Recognition (Eigenfaces)

Grayscale 2×2 face images (vectorized to 4D):

$$X = \begin{pmatrix} 200 & 180 & 190 & 185 \\ 180 & 200 & 185 & 195 \\ 80 & 90 & 85 & 88 \\ 20 & 30 & 25 & 28 \end{pmatrix}$$

(rows are pixels, columns are different faces)

- (a) Compute mean face $\bar{x}$ and center data. Find eigenfaces (principal components).
- (b) Represent a new face $f_{\text{new}} = [195, 188, 82, 22]^T$ using top 2 eigenfaces.
- (c) Derive the reconstruction and compare with original. Calculate mean squared error.

PROBLEM 15: Probabilistic PCA

Observed data: $x \in \mathbb{R}^d$ generated from latent variable $z \in \mathbb{R}^q$ where $q < d$.

Model: $x = Wz + \mu + \epsilon$ where $z \sim \mathcal{N}(0, I)$, $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$.

- (a) Derive the marginal distribution: $x \sim \mathcal{N}(\mu, WW^T + \sigma^2 I)$.
- (b) Show that maximum likelihood estimate of W gives: $W_{ML} = V_q(\Lambda_q - \sigma^2 I)^{1/2} R$ where V_q contains top q eigenvectors and R is arbitrary rotation.
- (c) Prove that as $\sigma^2 \rightarrow 0$, probabilistic PCA reduces to standard PCA. What's the advantage of the probabilistic formulation?

Gaussian Distribution

PROBLEM 16: Quality Control with Normal Distribution

Manufacturing process produces bolts with diameter $\sim \mathcal{N}(10, 0.04)$ mm. Specifications: 9.7–10.3 mm.

- (a) Calculate probability a random bolt meets specifications. Use error function: $\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$.
- (b) If 10,000 bolts are produced daily, how many defective bolts are expected?
- (c) Derive the process capability index $C_p = \frac{USL-LSL}{6\sigma}$. If $C_p < 1.33$, what σ is needed?

PROBLEM 17: Multivariate Gaussian Conditional Distribution

Bivariate distribution: height X_1 and weight X_2 with

$$\boldsymbol{\mu} = \begin{pmatrix} 170 \\ 70 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} 64 & 32 \\ 32 & 25 \end{pmatrix}$$

- (a) Derive the conditional distribution $X_2|X_1 = 180$ using the formula:

$$\mu_{2|1} = \mu_2 + \Sigma_{21}\Sigma_{11}^{-1}(x_1 - \mu_1), \quad \Sigma_{2|1} = \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}$$

- (b) Calculate expected weight for height = 180 cm and the conditional variance.
- (c) Prove the conditional distribution is also Gaussian. Why is this property important in Bayesian inference?

PROBLEM 18: Chi-Square Goodness-of-Fit Test

Dice rolled 600 times with observed frequencies:

Face	1	2	3	4	5	6
Observed	92	107	98	95	103	105

- (a) Test H_0 : dice is fair using $\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$. Calculate test statistic.
- (b) Find critical value for $\alpha = 0.05$ with appropriate df. Make decision about fairness.
- (c) Derive the asymptotic distribution of χ^2 statistic using central limit theorem. Why does it follow chi-square distribution?

PROBLEM 19: Gaussian Mixture Model Classification

Two classes with Gaussian distributions:

- Class 1: $\mathcal{N}(2, 1)$, prior $\pi_1 = 0.6$
- Class 2: $\mathcal{N}(5, 1.5)$, prior $\pi_2 = 0.4$

- (a) Derive Bayes optimal decision boundary by solving: $\pi_1 p(x|C_1) = \pi_2 p(x|C_2)$.
- (b) Classify observation $x = 3.5$ using posterior probabilities: $P(C_i|x) = \frac{\pi_i p(x|C_i)}{\sum_j \pi_j p(x|C_j)}$.
- (c) Calculate expected misclassification rate for this decision rule.

PROBLEM 20: Q-Q Plot for Normality Testing

Sample data (sorted): $\{12.3, 14.1, 15.8, 16.2, 17.5, 18.9, 20.1, 22.4\}$.

- (a) Calculate theoretical quantiles from $\mathcal{N}(0, 1)$ for plotting positions $p_i = \frac{i-0.5}{n}$. Use $\Phi^{-1}(p_i)$.
- (b) Standardize data: $z_i = \frac{x_i - \bar{x}}{s}$. Plot $(z_{(i)}, q_i)$ and assess linearity.
- (c) Perform Shapiro-Wilk test using: $W = \frac{(\sum a_i x_{(i)})^2}{\sum (x_i - \bar{x})^2}$. Interpret result (W close to 1 indicates normality).

Ridge Regression

PROBLEM 21: Housing Price Prediction with Multicollinearity

Predicting house price from size (x_1) and rooms (x_2) (size and rooms are highly correlated):

$$X = \begin{pmatrix} 1 & 1000 & 2 \\ 1 & 1500 & 3 \\ 1 & 2000 & 4 \\ 1 & 2500 & 5 \end{pmatrix}, \quad y = \begin{pmatrix} 200 \\ 270 \\ 340 \\ 410 \end{pmatrix}$$

(prices in thousands)

- (a) Compute $X^T X$ and calculate its condition number $\kappa = \lambda_{\max}/\lambda_{\min}$. What does high κ indicate?
- (b) Derive and compute ridge solution with $\lambda = 100$: $\hat{\beta}^{\text{ridge}} = (X^T X + \lambda I)^{-1} X^T y$.
- (c) Compare with OLS. Show how ridge shrinks coefficients and reduces variance at cost of bias.

PROBLEM 22: Cross-Validation for Lambda Selection

Small dataset: $X = \begin{pmatrix} 1 & 2 \\ 1 & 4 \\ 1 & 6 \\ 1 & 8 \end{pmatrix}, y = \begin{pmatrix} 3 \\ 7 \\ 9 \\ 13 \end{pmatrix}.$

- (a) Implement leave-one-out cross-validation for $\lambda \in \{0, 1, 5, 10\}$. Calculate CV error: $\text{CV}(\lambda) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_{-i}^{(\lambda)})^2$.
- (b) Derive the LOOCV shortcut formula: $\text{CV}(\lambda) = \frac{1}{n} \sum \left(\frac{y_i - \hat{y}_i}{1 - H_{ii}} \right)^2$ where $H = X(X^T X + \lambda I)^{-1} X^T$.
- (c) Plot CV error vs λ . Which λ is optimal? Explain bias-variance trade-off.

PROBLEM 23: Ridge Regression as Constrained Optimization

Consider equivalent formulations of ridge regression.

- (a) Show that ridge regression $\min_{\beta} \|y - X\beta\|^2 + \lambda \|\beta\|^2$ is equivalent to constrained problem: $\min_{\beta} \|y - X\beta\|^2$ subject to $\|\beta\|^2 \leq t$ for some $t(\lambda)$.
- (b) Derive the Lagrangian and KKT conditions. Find the relationship between λ and constraint t .
- (c) Geometrically illustrate ridge solution as intersection of constraint ball $\|\beta\| \leq \sqrt{t}$ and OLS error ellipsoids. Why does ridge shrink towards origin?

PROBLEM 24: Generalized Cross-Validation (GCV)

For design matrix X and penalty λ .

- (a) Derive the GCV criterion: $\text{GCV}(\lambda) = \frac{n\|y - X\hat{\beta}^\lambda\|^2}{(n - \text{df}(\lambda))^2}$ where $\text{df}(\lambda) = \text{Tr}[X(X^T X + \lambda I)^{-1} X^T]$.
- (b) For dataset: $X = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}$, $y = \begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix}$, compute $\text{GCV}(\lambda)$ for $\lambda = 0.5$.
- (c) Explain why GCV approximates leave-one-out CV but is computationally cheaper.

PROBLEM 25: Tikhonov Regularization (Generalized Ridge)

Generalized ridge with penalty matrix Γ : $\min_{\beta} \|y - X\beta\|^2 + \lambda\|\Gamma\beta\|^2$.

- (a) Derive solution: $\hat{\beta} = (X^T X + \lambda\Gamma^T \Gamma)^{-1} X^T y$. For $\Gamma = D$ (difference matrix), what constraints are imposed?
- (b) Consider trend filtering with $D = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}$ (first differences). Solve for data:
 $X = I_3$, $y = \begin{pmatrix} 1 \\ 5 \\ 2 \end{pmatrix}$, $\lambda = 1$.
- (c) Relate Tikhonov regularization to prior $\beta \sim \mathcal{N}(0, (\lambda\Gamma^T \Gamma)^{-1})$. What does $\Gamma = D$ encode?

Controlling Standard Deviation

PROBLEM 26: Pharmaceutical Tablet Weight Variance

Tablet weight regulation requires $\sigma \leq 2$ mg. Sample of 25 tablets has $s = 2.3$ mg.

- (a) Test $H_0 : \sigma^2 = 4$ vs $H_a : \sigma^2 > 4$ at $\alpha = 0.05$ using $\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$. Critical value: $\chi_{0.05,24}^2 = 36.42$.
- (b) Calculate the p-value: $P(\chi_{24}^2 > \chi_{\text{obs}}^2)$. Make regulatory decision.
- (c) Derive power of test: $1 - \beta = P(\text{reject } H_0 | \sigma^2 = 6)$. How many samples needed for 80% power?

PROBLEM 27: Comparing Process Variabilities

Two assembly lines produce resistors. Line A: $n_1 = 16, s_1^2 = 0.49$. Line B: $n_2 = 21, s_2^2 = 0.25$.

- (a) Test equality of variances: $H_0 : \sigma_1^2 = \sigma_2^2$ using $F = s_1^2/s_2^2 \sim F_{15,20}$. Critical values: $F_{0.025,15,20} = 2.76, F_{0.975,15,20} = 0.39$.
- (b) Construct 95% CI for variance ratio: $\left[\frac{s_1^2}{s_2^2} \cdot \frac{1}{F_{\alpha/2}}, \frac{s_1^2}{s_2^2} \cdot \frac{1}{F_{1-\alpha/2}} \right]$.
- (c) If variances differ significantly, which statistical tests become invalid? Suggest robust alternatives.

PROBLEM 28: Bartlett's Test for Homogeneity of Variances

Three groups with sample sizes $n = (10, 12, 15)$ and variances $s^2 = (4.2, 3.8, 5.1)$.

- (a) Calculate pooled variance: $s_p^2 = \frac{\sum (n_i - 1)s_i^2}{\sum (n_i - 1)}$.
- (b) Compute Bartlett's statistic: $B = \frac{(N-k) \ln(s_p^2) - \sum (n_i - 1) \ln(s_i^2)}{1 + \frac{1}{3(k-1)} \left[\sum \frac{1}{n_i - 1} - \frac{1}{N - k} \right]}$ where $N = \sum n_i, k = 3$.
- (c) Under H_0 : equal variances, $B \sim \chi_{k-1}^2$. Test at $\alpha = 0.05$ with $\chi_{0.05,2}^2 = 5.99$.

PROBLEM 29: Brown-Forsythe Test (Robust to Non-Normality)

Data from 3 groups (deviations from median):

- Group 1: $|x_i - \text{med}_1| = \{2.1, 1.8, 2.3, 1.9\}$
- Group 2: $|x_i - \text{med}_2| = \{3.2, 2.9, 3.5, 3.1\}$
- Group 3: $|x_i - \text{med}_3| = \{1.5, 1.7, 1.4, 1.6\}$

- (a) Perform ANOVA on absolute deviations to test variance equality. Calculate $F = \frac{MS_{\text{between}}}{MS_{\text{within}}}$.

- (b) Why is Brown-Forsythe preferred over Bartlett's for non-normal data? Explain robustness.
- (c) If H_0 rejected, perform post-hoc pairwise comparisons using Bonferroni correction. Which groups differ?

PROBLEM 30: Cochran's C Test for Outlier Variance

Experiment conducted 6 times with sample variances: $s^2 = \{4.1, 3.9, 4.2, 3.8, 4.0, 12.5\}$ (all $n_i = 5$).

- (a) Calculate Cochran's statistic: $C = \frac{\max(s_i^2)}{\sum s_i^2} = \frac{12.5}{32.5}$.
- (b) Compare with critical value $C_{0.05}(6, 4) = 0.616$ (6 groups, $df = 4$ each). Is largest variance an outlier?
- (c) If outlier detected, investigate: run Grubbs' test on original data from that group. Derive decision rule.

Markov Chains

PROBLEM 31: Brand Loyalty Model

Customers switch between brands A, B, C monthly:

$$P = \begin{pmatrix} 0.7 & 0.2 & 0.1 \\ 0.3 & 0.6 & 0.1 \\ 0.2 & 0.3 & 0.5 \end{pmatrix}$$

Current market shares: $\pi^{(0)} = (0.4, 0.35, 0.25)$.

- (a) Calculate market shares after 3 months: $\pi^{(3)} = \pi^{(0)}P^3$. Show intermediate steps.
- (b) Find equilibrium market shares by solving $\pi P = \pi$, $\sum \pi_i = 1$. Use substitution or eigenvalue method.
- (c) Derive mean return time to brand A: $\tau_A = 1/\pi_A$. Interpret in business context.

PROBLEM 32: Absorbing Markov Chain - Credit Rating

Bond ratings: AAA, AA, A, D (default, absorbing). Transition matrix:

$$P = \begin{pmatrix} 0.90 & 0.08 & 0.02 & 0 \\ 0.05 & 0.85 & 0.08 & 0.02 \\ 0.02 & 0.10 & 0.80 & 0.08 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

- (a) Identify transient and absorbing states. Write canonical form: $P = \begin{pmatrix} Q & R \\ 0 & I \end{pmatrix}$.
- (b) Calculate fundamental matrix $N = (I - Q)^{-1}$. Interpret N_{ij} as expected visits to state j starting from i .
- (c) Find absorption probabilities: $B = NR$. What's probability an AA bond eventually defaults?

PROBLEM 33: PageRank Algorithm

Web graph with 4 pages and link structure:

$$L = \begin{pmatrix} 0 & 1/2 & 1/2 & 0 \\ 1/3 & 0 & 1/3 & 1/3 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

- (a) Form Google matrix: $G = \alpha L + (1 - \alpha) \frac{\mathbf{1}\mathbf{1}^T}{n}$ with $\alpha = 0.85$. Why is teleportation term needed?

- (b) Compute PageRank by power iteration: $\pi^{(k+1)} = G^T \pi^{(k)}$ starting from $\pi^{(0)} = (0.25, 0.25, 0.25, 0.25)$. Iterate until $\|\pi^{(k+1)} - \pi^{(k)}\| < 10^{-3}$.
- (c) Prove convergence using Perron-Frobenius theorem. What conditions ensure unique stationary distribution?

PROBLEM 34: Birth-Death Process for Queuing

Server with arrival rate $\lambda = 3$ customers/hr, service rate $\mu = 4$ customers/hr.

Generator matrix for states $n = 0, 1, 2, 3$ (max capacity):

$$Q = \begin{pmatrix} -3 & 3 & 0 & 0 \\ 4 & -7 & 3 & 0 \\ 0 & 4 & -7 & 3 \\ 0 & 0 & 4 & -4 \end{pmatrix}$$

- (a) Verify detailed balance: $\pi_i q_{ij} = \pi_j q_{ji}$. Derive stationary distribution $\pi_n = \pi_0 (\lambda/\mu)^n$.
- (b) Calculate average queue length: $L = \sum n \pi_n$ and expected wait time using Little's law: $W = L/\lambda$.
- (c) Find probability of system being full (customer rejected). How does this change if μ increases to 5?

PROBLEM 35: Hidden Markov Model Basics

2-state HMM: Sunny (S) and Rainy (R) with transitions and emissions.

Transition: $A = \begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix}$. Emission: $B = \begin{pmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{pmatrix}$ (Happy/Grumpy).

Observe: Happy, Happy, Grumpy.

- (a) Calculate $P(\text{observations} | \text{states} = SSR)$ using $P(O|\lambda) = \prod P(o_t | s_t) \cdot P(s_t | s_{t-1})$ with initial $\pi = (0.6, 0.4)$.
- (b) Implement forward algorithm to compute $P(\text{observations})$: $\alpha_t(i) = P(o_1, \dots, o_t, s_t = i)$.
- (c) Use Viterbi algorithm to find most likely state sequence. Show trellis diagram.

Monte Carlo Methods

PROBLEM 36: Buffon's Needle for Estimating Pi

Needle of length $L = 1$ dropped on floor with parallel lines distance $D = 2$ apart.

- (a) Derive probability needle crosses a line: $P = \frac{2L}{\pi D}$. (Hint: parameterize by angle θ and distance x from nearest line).
- (b) Simulate 10,000 drops. Estimate π using $\hat{\pi} = \frac{2L \cdot n_{\text{total}}}{D \cdot n_{\text{cross}}}$. Calculate standard error.
- (c) Compare efficiency with circle-in-square method. Which has lower variance?

PROBLEM 37: Stratified Sampling for Variance Reduction

Estimate $I = \int_0^4 x^3 dx$ (true value = 64).

- (a) Standard MC with $n = 100$ uniform samples on $[0, 4]$: $\hat{I}_{\text{MC}} = 4 \cdot \frac{1}{n} \sum f(U_i)$. Calculate variance.
- (b) Stratified sampling: divide $[0, 4]$ into 4 equal strata. Sample 25 points per stratum. Calculate $\hat{I}_{\text{strat}} = \sum w_k \bar{f}_k$ where $w_k = 0.25$.
- (c) Prove variance reduction: $\text{Var}(\hat{I}_{\text{strat}}) = \sum w_k^2 \sigma_k^2 / n_k < \text{Var}(\hat{I}_{\text{MC}})$. Verify numerically.

PROBLEM 38: Control Variates Method

Estimate $\theta = E[e^X]$ where $X \sim \text{Uniform}(0, 1)$.

Control variate: $Y = X$ with known $E[Y] = 0.5$.

- (a) Simulate $n = 1000$ samples. Estimate $\hat{\theta}_0 = \frac{1}{n} \sum e^{X_i}$ and calculate sample variance.
- (b) Use control variate estimator: $\hat{\theta}_c = \hat{\theta}_0 - c(\bar{Y} - 0.5)$ where $c = \text{Cov}(e^X, X) / \text{Var}(X)$. Estimate c from sample covariance.
- (c) Show variance reduction: $\text{Var}(\hat{\theta}_c) = \text{Var}(\hat{\theta}_0)(1 - \rho^2)$ where $\rho = \text{Cor}(e^X, X)$. Calculate efficiency gain.

PROBLEM 39: Gibbs Sampling for Bivariate Normal

Target: $(X, Y) \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix}\right)$.

Conditionals: $X|Y \sim \mathcal{N}(0.8Y, 0.36)$, $Y|X \sim \mathcal{N}(0.8X, 0.36)$.

- (a) Implement Gibbs sampler for 5000 iterations starting at $(X_0, Y_0) = (0, 0)$. Plot trace of X_t .

- (b) Calculate autocorrelation: $\rho_k = \text{Cor}(X_t, X_{t+k})$ for lags $k = 1, \dots, 50$. Estimate effective sample size: $n_{\text{eff}} = \frac{n}{1+2\sum \rho_k}$.
- (c) Compare marginal histograms with true $\mathcal{N}(0, 1)$. Perform Kolmogorov-Smirnov test for goodness of fit.

PROBLEM 40: Rejection Sampling from Truncated Distribution

Target: truncated normal $X \sim \mathcal{N}(0, 1)$ on $[2, \infty)$.

Proposal: exponential $g(x) = e^{-(x-2)}$ for $x \geq 2$.

- (a) Find constant M such that $f(x) \leq M g(x)$ for all $x \geq 2$ where $f(x) = \frac{\phi(x)}{\Phi(-2)}$.
- (b) Implement rejection algorithm:
 - (a) Generate $Y \sim g$, $U \sim \text{Uniform}(0, 1)$
 - (b) Accept if $U \leq f(Y)/(M g(Y))$

Run until 1000 samples accepted. Calculate acceptance rate.

- (c) Derive expected number of proposals per acceptance: $1/P_{\text{accept}} = M$. Discuss efficiency vs direct sampling.

Deterministic Models

PROBLEM 41: Pharmacokinetics - Drug Concentration Model

One-compartment model: $\frac{dC}{dt} = -kC$ where $C(t)$ is drug concentration, $k = 0.2 \text{ hr}^{-1}$.

Single dose: $C(0) = 100 \text{ mg/L}$. Repeated dosing: $D = 50 \text{ mg/L}$ every 4 hours.

- (a) Solve ODE for single dose: $C(t) = C_0 e^{-kt}$. Calculate half-life: $t_{1/2} = \ln(2)/k$.
- (b) For repeated dosing, derive steady-state concentration: $C_{ss} = \frac{D}{1 - e^{-kT}}$ where $T = 4 \text{ hr}$ is dosing interval. Calculate C_{ss} .
- (c) Find therapeutic window $[C_{\min}, C_{\max}] = [20, 80]$. Adjust D and T to maintain $C(t)$ in this range.

PROBLEM 42: Chemical Reaction Kinetics

Second-order reaction: $A + B \rightarrow C$ with rate law $\frac{d[A]}{dt} = -k[A][B]$.

Initial: $[A]_0 = [B]_0 = 1.0 \text{ M}$, $k = 0.5 \text{ M}^{-1}\text{min}^{-1}$.

- (a) For equal initial concentrations, derive: $\frac{1}{[A]} - \frac{1}{[A]_0} = kt$. Solve for $[A](t)$.
- (b) Calculate time for 75% conversion ($[A] = 0.25 \text{ M}$). At what time does $[A] = 0.1[A]_0$?
- (c) If $[A]_0 \neq [B]_0$, derive: $\ln \frac{[A][B]_0}{[B][A]_0} = ([B]_0 - [A]_0)kt$. Solve for $[A]_0 = 1.0$, $[B]_0 = 2.0$.

PROBLEM 43: Zombie Apocalypse Model (SZR)

Population dynamics:

$$\begin{cases} \frac{dS}{dt} = -\beta SZ \\ \frac{dZ}{dt} = \beta SZ + \zeta R - \alpha SZ \\ \frac{dR}{dt} = \alpha SZ - \zeta R \end{cases}$$

Parameters: $\beta = 0.0001$ (infection), $\alpha = 0.0001$ (kill), $\zeta = 0.01$ (resurrection). Initial: $S(0) = 1000$, $Z(0) = 10$, $R(0) = 0$.

- (a) Verify conservation when $\zeta = 0$: $S + Z + R = N$. What happens with $\zeta > 0$?
- (b) Simulate for 100 days using Euler method with $\Delta t = 0.1$. Plot $S(t)$, $Z(t)$, $R(t)$.
- (c) Find zombie-free equilibrium. Derive stability condition (zombie extinction threshold).

PROBLEM 44: Van der Pol Oscillator (Nonlinear Dynamics)

$$\frac{d^2x}{dt^2} - \mu(1 - x^2)\frac{dx}{dt} + x = 0$$

Rewrite as system: $\frac{dx}{dt} = y$, $\frac{dy}{dt} = \mu(1 - x^2)y - x$.

- (a) Find fixed points. For $\mu = 1$, analyze stability using Jacobian at origin.
- (b) Simulate with initial conditions $(x_0, y_0) = (0.1, 0)$ for $\mu \in \{0.1, 1, 5\}$ over $t \in [0, 50]$. Plot phase portraits.
- (c) Explain limit cycle behavior. How does period depend on μ ? (numerical approximation acceptable)

PROBLEM 45: Optimal Control - Vaccination Strategy

SIR model with vaccination: control $u(t) \in [0, 0.9]$ (fraction vaccinated daily).

$$\begin{cases} \frac{dS}{dt} = -\beta SI - u(t)S \\ \frac{dI}{dt} = \beta SI - \gamma I \\ \frac{dR}{dt} = \gamma I + u(t)S \end{cases}$$

Objective: minimize $J = \int_0^T [\alpha I(t) + \beta u(t)^2] dt$ (balance infection cost vs vaccination cost).

Parameters: $\beta = 0.0005, \gamma = 0.1, \alpha = 1, \beta = 10, T = 100$.

- (a) Write Hamiltonian: $H = \alpha I + \beta u^2 + \lambda_S(-\beta SI - uS) + \lambda_I(\beta SI - \gamma I) + \lambda_R(\gamma I + uS)$.
- (b) Derive optimal control from $\frac{\partial H}{\partial u} = 0$: $u^* = \max\{0, \min\{0.9, -\lambda_S S / (2\beta)\}\}$.
- (c) Implement forward-backward sweep method to solve. Compare total cost with constant vaccination $u = 0.3$.